DEPARTMENT OF ELECTRICAL
AND COMPUTER ENGINEERING

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BSc in Electrical and Computer Engineering

5G NETWORKS: DEVELOPMENT OF A 5G CORE AND RADIO ACCESS NETWORK

PRE-DISSERTATION INFORMATION

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NOVA University Lisbon

Draft: *May 2, 2026*

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Listings

Acronyms

COTS Commercial Off-The-Shelf *(p. 3)*

eNodeB Evolved Node B *(p. 3)*

EPC Evolved Packet Core *(pp. 2, 3, 5)*

RAN Radio Access Network *(pp. 2–4)*

SDR Software-Defined Radio *(pp. 3, 4)*

UE User Equipment *(p. 3)*

Bibliography

[1] *MongoDB Community Edition: Installation Guide for Linux*. URL: https://www.mongodb.com/docs/manual/administration/install-community/?operating-system=linux&linux-distribution=ubuntu&linux-package=default&search-linux=with-search-linux (cit. on p. 6).

[2] *Open5GS Guide: Quick Start*. URL: https://open5gs.org/open5gs/docs/guide/01-quickstart/ (cit. on p. 5).

4G Network Setup

Before developing the 5G network testbed, a 4G network was first established to serve as a performance baseline and a proof-of-concept for all hardware and software components. The 4G network setup was essential for several reasons:

1. Validating that the chosen hardware and software stack functioned reliably
2. Establishing a known and well-documented reference point for network behavior
3. Identifying potential deployment hurdles before scaling to 5G complexity

This appendix documents the complete 4G testbed architecture, deployment procedures, and validation results.

The network stack consists of two dedicated machines running the backbone infrastructure: one running Open5GS for the Evolved Packet Core (EPC), while the other runs srsRAN 4G for the Radio Access Network (RAN). Readers can use this section as a guide for replicating the network, and as a foundation for understanding the experimental setup that enabled the 5G network development described in the main thesis.

This section will provide a comprehensive guide on how the 4G network was established, including a description of the equipment used for the setup and the software chosen for the network stack, the necessary setup steps and specific configurations applied to optimize performance, and the testing methodologies employed to validate the network's functionality and performance.

As an effort to make the network as reproducible as possible, the setup process will be detailed step-by-step, starting from the initial hardware assembly to the final testing phase. This will include instructions on how to connect the various components, configure the network settings, and troubleshoot common issues that may arise during the setup process.

.1 System Architecture Overview

The 4G testbed is a self-contained LTE system comprising three main functional domains:

- **RAN**: Handles air-interface communication using srsRAN Evolved Node B (eNodeB) software running on a dedicated machine, along with a Nuand BladeRF Software-Defined Radio (SDR) and appropriate antennas for signal transmission and reception
- **EPC**: Implements the core network functions using Open5GS software running on a separate machine, providing essential services such as authentication, mobility management, and data routing to the internet
- **User Equipment (UE)**: Physical Commercial Off-The-Shelf (COTS) device configured to connect to the RAN and access the services provided by the EPC

Figure .1 presents the logical architecture.

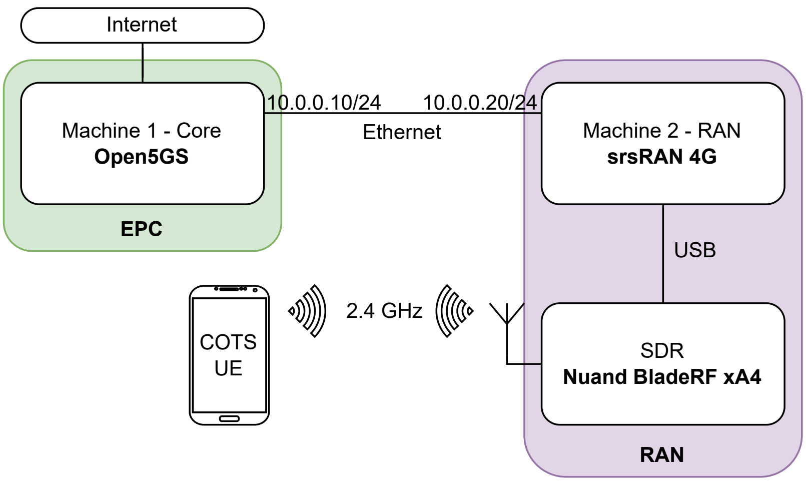

Figure .1: Logical architecture of the 4G network, illustrating the interactions between the RAN, EPC, and UE components.

.2 Hardware Components

The hardware used for the 4G network setup is composed of two identical machines, a Software-Defined Radio and antennas.

The computers running the 4G network are DELL Optiplex Micro 7020, with the following specifications:

Component	Machine 1	Machine 2
CPU	Intel Core i7-14700T	Intel Core i7-14700T
RAM	16 GB DDR5	16 GB DDR5
Operating System	Ubuntu 22.04 LTS	Ubuntu 22.04 LTS
Storage	512 GB SSD	512 GB SSD

Table .2: Specifications of the machines used for the 4G network setup.

Data flows between the machines via a dedicated Ethernet backhaul network. The RAN on Machine 2 establishes a connection to the EPC on Machine 1 using SCTP (Stream Control Transmission Protocol). When a UE initiates a connection, the RAN forwards authentication requests to the core, which validates the subscriber and allocates IP resources. All user data packets then traverse through the core network functions before reaching their destination.

The radio interface of the 4G network is implemented using a Nuand BladeRF xA4 SDR. This device connects the Machine 2 to provide the necessary radio capabilities for the RAN. In practical terms, the host machine executes the signal processing and network stack components, whereas the BladeRF xA4 manages the actual radio I/O. This separation of responsibilities improves flexibility, simplifies experimentation, and makes the overall testbed easier to adapt to different deployment scenarios.

Below, in Figure .2, shows the physical setup of the 4G network, including the two machines, the SDR, and the antennas. The photograph illustrates how the components are arranged in the lab environment, providing a visual reference for replicating the setup.

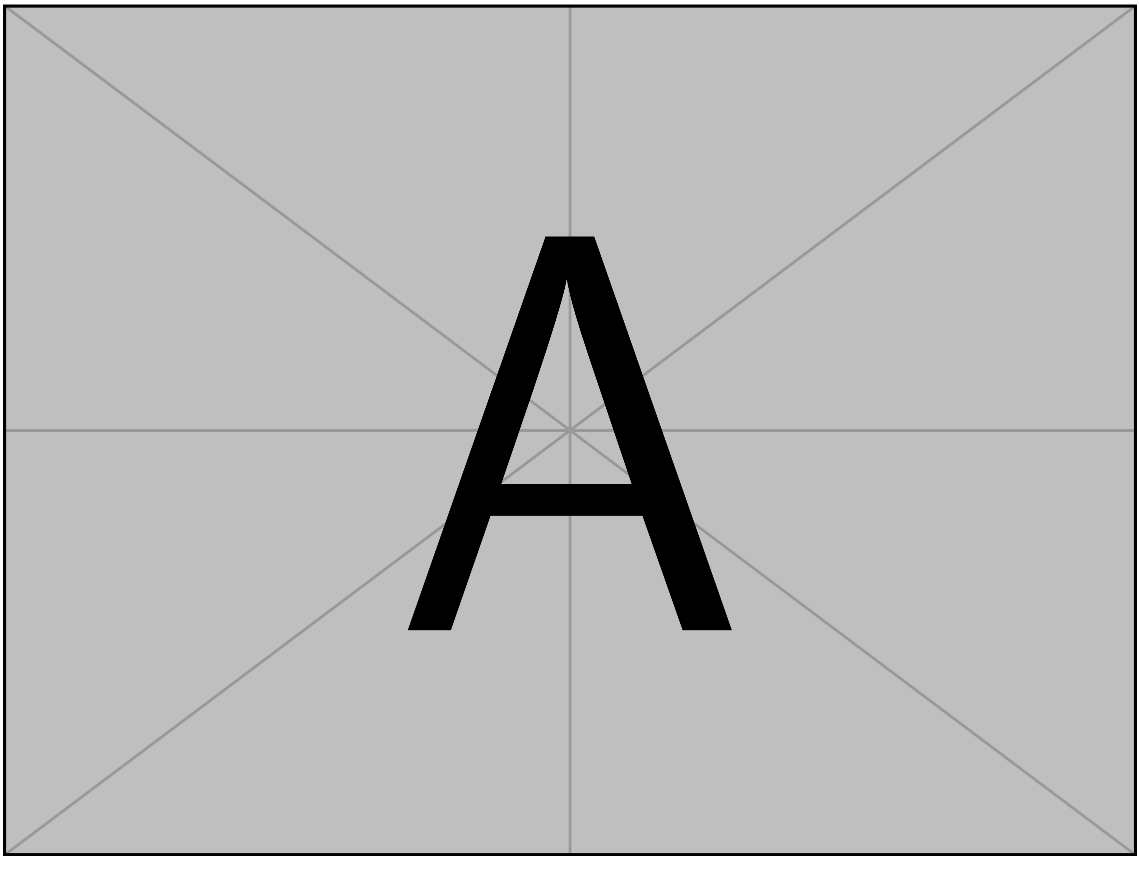

Figure .2: Photograph of the 4G network setup, showing the two machines, the Nuand BladeRF xA4 SDR, and the antennas used for the radio interface.

.3 Software Components

.4 Software Setup

.4.1 Open5GS

This section provides a step-by-step guide for installing and configuring Open5GS. The installation process comprises of the following steps:

1. Installing MongoDB.
2. Adding the Open5GS repository and installing Open5GS packages.
3. Installing the WebUI for Open5GS.
4. Configuring Open5GS components (MME, HSS, SGW, PGW) according to the network requirements.

The necessary steps and more information about the project and installation steps can be found in the official Open5GS documentation [2]. Unless otherwise specified, all steps should be executed on the machine designated to run the EPC (Machine 1 in our setup).

.4.1.1 Installing MongoDB

Open5GS depends on MongoDB as its database backend, where it stores subscriber information, network configurations, and operational data. Therefore, the first step in setting up Open5GS is to install MongoDB:

```
sudo apt update
sudo apt install gnupg curl -y
curl -fsSL https://pgp.mongodb.com/server-8.0.asc | \
  sudo gpg -o /usr/share/keyrings/mongodb-server-8.0.gpg --dearmor

echo "deb [ arch=amd64,arm64 signed-by=/usr/share/keyrings/\
    mongodb-server-8.0.gpg ] https://repo.mongodb.com/apt/ubuntu \
    jammy/mongodb-enterprise/8.2 multiverse" | sudo tee \
    /etc/apt/sources.list.d/mongodb-enterprise-8.2.list

sudo apt update
sudo apt-get install -y mongodb-org
```

After executing the above commands, MongoDB will be installed on the machine. You can verify the installation by checking the MongoDB service status:

```
sudo systemctl status mongod
```

In which the output should indicate that the MongoDB service is active and running:

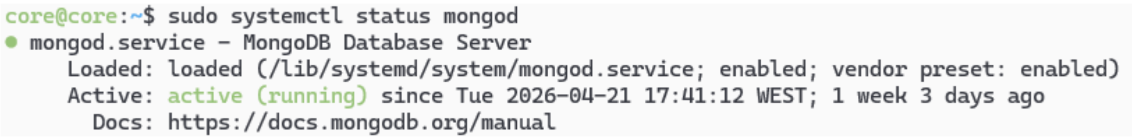

Figure .3: Output of the command to check MongoDB service status, showing that the service is active and running.

To ensure MongoDB starts automatically on system boot, you can enable the service with the following command:

```
sudo systemctl enable mongod
```

For more detailed instructions and troubleshooting guidance, refer to the official MongoDB installation documentation [1].

.4.1.2 Adding Open5GS Repository and Installing Open5GS Packages

After installing MongoDB, the next step is to add the Open5GS repository to the system and install the necessary Open5GS packages. This can be done by executing the following commands:

```
sudo add-apt-repository ppa:open5gs/latest
sudo apt update
sudo apt install open5gs
```

This will install the Open5GS packages, including the core network functions such as MME, HSS, SGW, and PGW. After the installation is complete, you can verify that the Open5GS services are running correctly by checking their status:

```
sudo systemctl status open5gs-*
```

.4.1.3 Installing WebUI for Open5GS

.4.1.4 Configuring Open5GS Components